Dynamic Memory Allocation

TN

The process of allocating memory at run time is known as dynamic memory allocation. Dynamic memory management techniques allow us to allocate additional memory space or to release unwanted space at run time, thus, optimizing the use of storage space. Dynamic data structures provide flexibility in adding, deleting or rearranging data items at run time.

In C, there are 4 library functions known as "memory management functions" that can be used for allocating and freeing memory during program execution.

1) **malloc() function :-**

This function reserves a block of memory of specified size and returns a pointer of type void.

The general form is :-

```
ptr = (cast-type *) malloc (byte-size);
```

where ptr is a pointer of type "cast-type". The malloc() function returns a pointer (of cast-type) to an area of memory with size "byte-size".

The name malloc stands for "memory allocation".

P.T.O

Scanned by CamScanner

Example:-

```
x=(int *)malloc(100 * sizeof(int));
```

On successful execution of this statement, a memory space equivalent to "100 times the size of int" i.e. 200 bytes is reserved and the address of the first byte of the allocated memory is assigned to the pointer x of type int.

2) calloc() function:-

The difference between malloc() & calloc() is that while malloc() allocates a single block of storage space, calloc() allocates multiple blocks of storage, each of the same size, and then sets all bytes to zero.

The general form of calloc() is:-

```
ptr=(cast-type *)calloc(n, elem-size);
```

This statement allocates contiguous space for n blocks, each of size elem-size bytes. All bytes are initialized to zero and a pointer to the first byte of the allocated region is returned. If there is not enough space, a NULL pointer is returned.

Example:-

```
ptr=(float *) calloc (25, sizeof(float));
```

This statement allocates contiguous space in memory for an array of 25 elements each of size of float i.e. 4 bytes.

The name calloc stands for "contiguous allocation".

3) free() function:-

When we no longer need the data we stored in a block of memory, we may release that block of memory for future use, using the free() function:-

```
free(ptr);
```

where ptr is a pointer to a memory block, which has already been created by malloc() or calloc().

Scanned by CamScanner

4) realloc() function [TN]

It may so happen that the previously allocated memory is not sufficient and we need additional space for more elements. It is also possible that the memory allocated is much larger than necessary and we want to reduce it. In both cases, we can change the memory size already allocated with the function realloc(). For example, if the original allocation is done by the statement ptr = malloc(size);

then reallocation of space may be done by the statement

ptr = realloc (ptr, newsize);

This function allocates a new memory space of size "newsize" to ptr and returns a pointer to the first byte of the new memory block. "New size" may be larger or smaller than the original "size".

Q) Difference between Static Memory Allocation & Dynamic Memory Allocation.

Static Memory Allocation	Dynamic Memory Allocation
1) It refers to memory allocated at compile time.	1) It refers to memory allocated at run time.
2) The size of allocated memory is fixed when the program is created.	2) The size is not fixed when the program is created.
3) No special library functions are required for allocating memory.	3) 4 library functions are required for allocating and freeing memory — malloc(), calloc(), free() and realloc().
4) It may lead to wastage of memory since size is fixed at compile time.	4) Wastage of memory is avoided because memory is allocated at run time as per requirement.
5) Static memory is allocated in Stack.	5) Dynamic memory is allocated in heap.
6) Example:- int arr[20];	6) Example:- int *ptr; ptr = (int *)malloc(20 * sizeof(int));

Scanned by CamScanner

Q) Difference between malloc() & calloc().

Ans)

malloc()	calloc()
1) The name malloc() stands for memory allocation.	1) The name calloc() stands for contiguous allocation.
2) The general form of malloc() is:- ptr=(cast-type *)malloc(byte-size);	2) The general form of calloc() is:- ptr=(cast-type *)calloc(n, elem-size);
3) malloc() takes one argument which is number of bytes.	3) calloc() takes 2 arguments which are number of blocks and size of each block.
4) malloc() does not initialize the memory allocated.	4) calloc() initializes the allocated memory to zero.
5) malloc() is faster than calloc().	5) calloc() takes little longer than malloc() because of the extra step of initializing the allocated memory to zero.
6) Number of Bytes of memory are allocated by malloc().	6) ~~Blocks of~~ Number of blocks of memory are allocated by calloc().

Scanned by CamScanner